

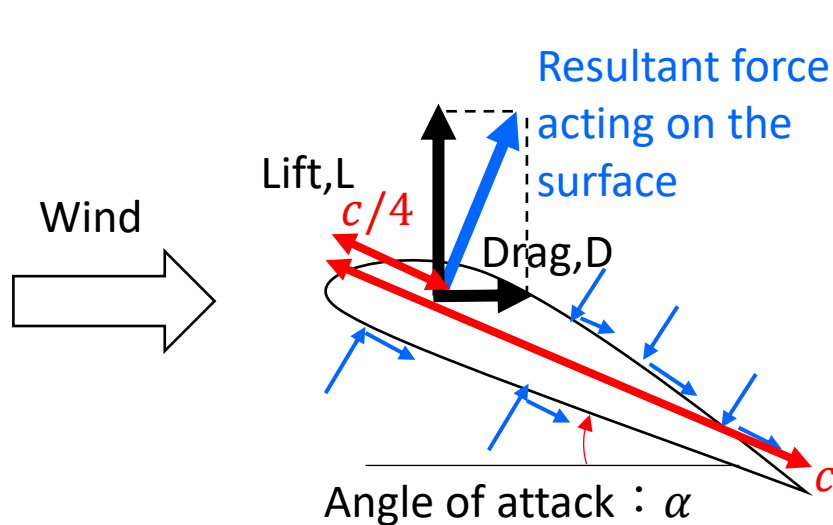
Aeronautical Systems B: Lecture 1

1. Aerodynamic Forces Acting on a Wing
2. Derivation of Lift Using Thin Airfoil Theory

Reference

Barnes W. McCormick, 'Aerodynamics, Aeronautics and Flight Mechanics', John Wiley & Sons, Inc., 1979

Aerodynamic Forces on a Wing



$$\text{Lift : } L = \frac{1}{2} \rho V^2 S C_L \quad (\text{B1-1})$$

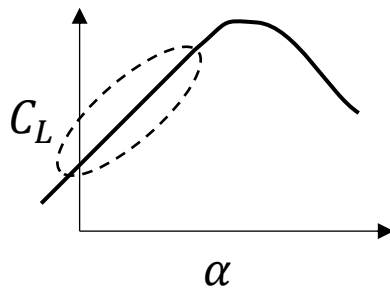
$$\text{Drag : } D = \frac{1}{2} \rho V^2 S C_D \quad (\text{B1-2})$$

ρ : Density V : Speed S : Area

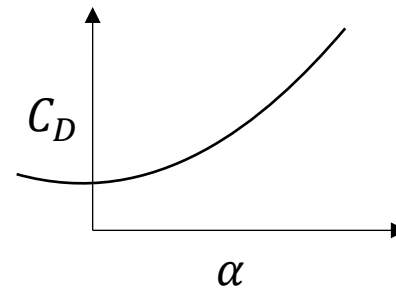
C_L : Lift coefficient C_D : Drag coefficient

$\frac{1}{2} \rho V^2$: Dynamic pressure

A force acts on a body surface due to the surrounding fluid (pressure acts normal to the surface and shear stress acts tangentially). The resultant of all forces acting on the body surface can be decomposed into components : the component normal to the wind is called lift (L), and the component parallel to the wind is called drag (D).



At low angles of attack,
slope of lift curve : $\frac{dC_L}{d\alpha} = \frac{2\pi}{c}$



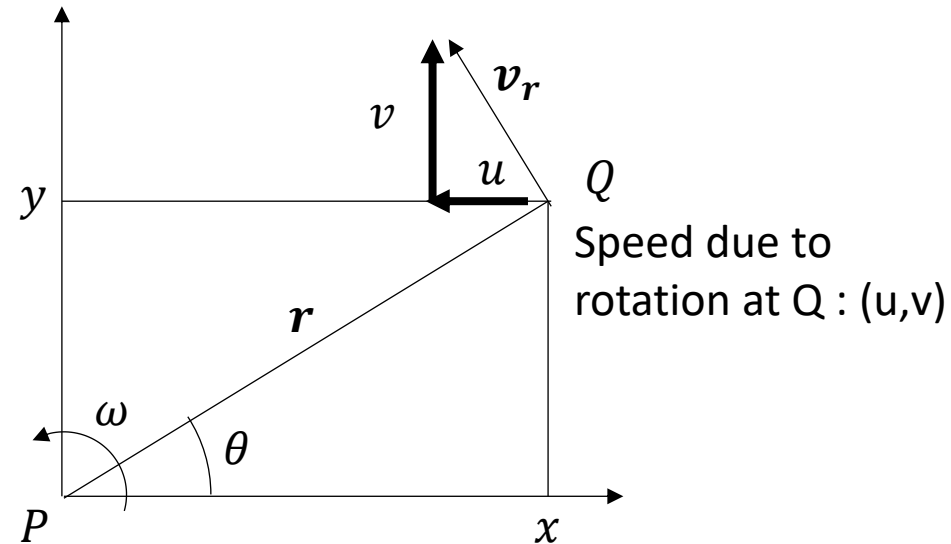
- For inviscid and incompressible flow, **drag = 0**.
- For a flat plate, the aerodynamic force acts at the **quarter-chord point**.

Vorticity

In a two-dimensional flow:

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (\text{B1-3})$$

Vorticity is **twice the angular velocity** of the fluid when it rotates as a rigid body.



Consider a fluid rotating locally as a rigid body about a point P. Let ω denote the angular velocity and v_r denote the radial velocity. The radial velocity due to rotation at point Q is given by:

$$v_r = \omega r \quad (\text{B1-4})$$

And,

$$u = -v_r \sin \theta = -\omega r \frac{y}{r} = -\omega y \quad (\text{B1-5})$$

$$v = v_r \cos \theta = \omega r \frac{x}{r} = \omega x \quad (\text{B1-6})$$

Taking partial derivatives with respect to y and x :

$$\frac{\partial u}{\partial y} = -\omega \quad (\text{B1-7})$$

$$\frac{\partial v}{\partial x} = \omega \quad (\text{B1-8})$$

Therefore,

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 2\omega \quad (\text{B1-9})$$

Circulation

$$\Gamma = \oint_c \mathbf{V} \cdot d\mathbf{s} \quad (\text{B1-10})$$

Line integral of velocity along a closed curve

Stokes' theorem

$$\Gamma = \iint_s \zeta \, dxdy \quad (\text{B1-11})$$

The area integral of vorticity is equal to the circulation along its boundary.

Biot-Savart law (2D)

$$v_i = \frac{\Gamma}{2\pi r} \quad (\text{B1-12})$$

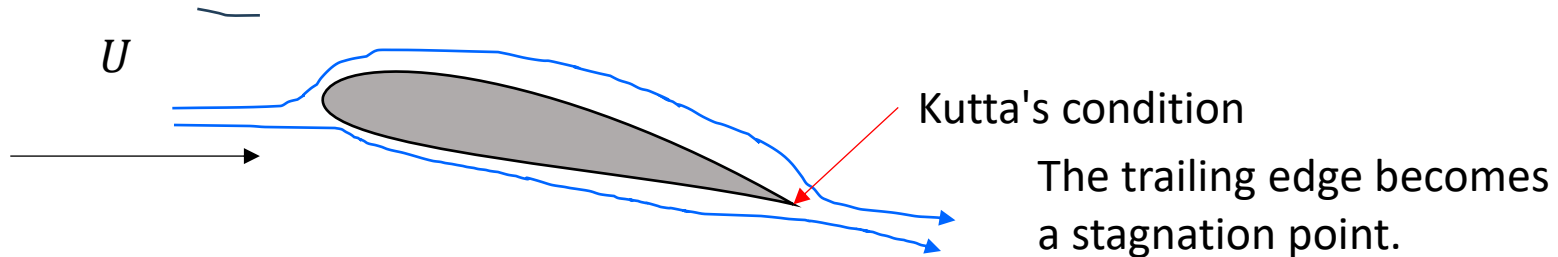
Velocity induced by a point vortex (potential vortex)

Kutta-Joukowski theorem

$$L = \rho U \Gamma \quad (\text{B1-13})$$

When an arbitrarily shaped two-dimensional object is placed in a uniform flow of a perfect fluid, the force acting on the object perpendicular to the flow is given by $\rho U \Gamma$, where ρ is the fluid density, U is the uniform flow velocity, and Γ is the circulation around the object.

We determine the lift acting on a thin airfoil placed in a uniform flow.



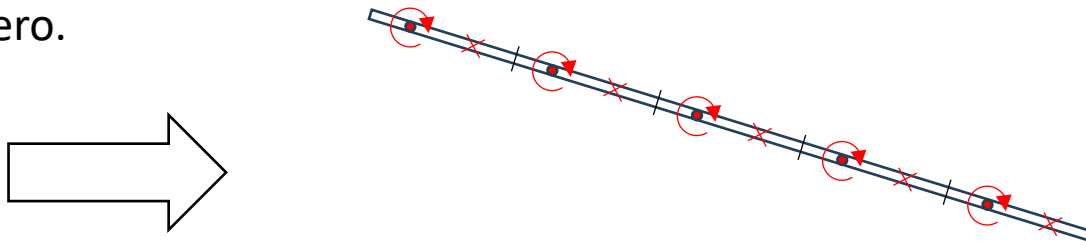
Weissinger's Approximation

(An approximate method for evaluating the aerodynamic characteristics of a thin airfoil)

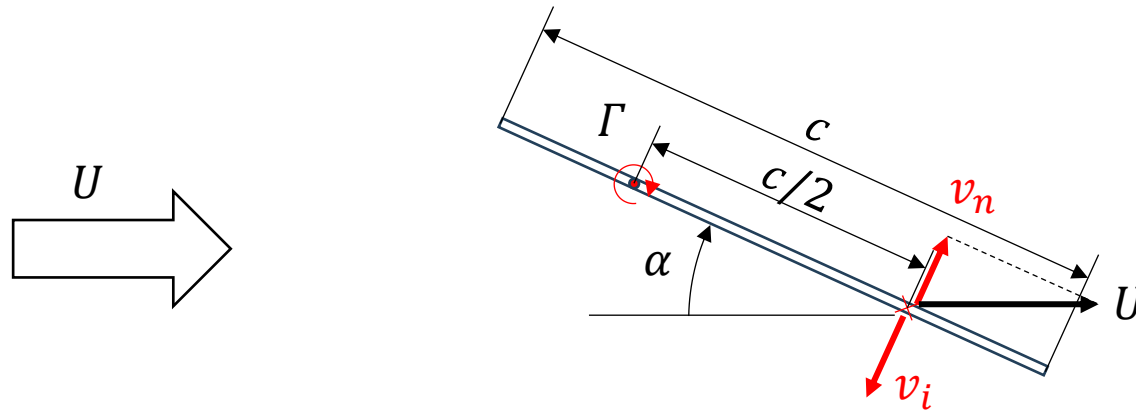
The airfoil chord is divided into several panels.

A vortex of unknown strength is placed at each **quarter-chord point** of these panels.

The unknown vortex strengths are determined from the condition that the **normal velocity component** at each **three-quarter-chord control point** becomes zero.



【Example】 Single-panel flat plate



Induced velocity at the 3/4 chord point due to vorticity U

$$v_i = \frac{\Gamma}{2\pi r}, \quad r = c/2 \quad (\text{B1-14})$$

Vertical component of the wing in uniform flow U

$$v_n = U \sin \alpha = U\alpha \quad (\text{B1-15})$$

$$v_i = v_n$$

Therefore,

$$\frac{\Gamma}{\pi c} = U\alpha \quad \Gamma = \pi c U\alpha \quad (\text{B1-16})$$

From the Kutta–Joukowski theorem, the lift is:

$$L = \rho \pi c U^2 \alpha \quad (\text{B1-17})$$

In non-dimensional form:

$$C_l = \frac{L}{\frac{1}{2} \rho U^2 c} = 2\pi \alpha \quad (\text{B1-18})$$

✂ For two-dimensional wings, use lowercase C_l instead of uppercase C_L

Lift-curve slope:

$$\frac{\partial C_l}{\partial \alpha} = 2\pi \quad (\text{B1-19})$$